Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/16/2018
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	January 2018 report: Prof. Alain Karma is no longer a Co-Investigator on this project, but continues to be a collaborator, who is responsible for the modeling part of the project through a separate NASA grant.		
COI Name (Institution):			
Grant/Contract No.:	NNX16AL77G		
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	NOTE: Continuation of "Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)" grant number NNX12AK54G, with the same Principal Investigator Dr. Rohit Trivedi. Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic objects produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena, and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime.		

Task Description:	While extensive experimental and theoretical work during the past several decades has yielded basic insights into complex solidification patterns, their dynamics in three dimensions (3D) still remains poorly characterized and understood. Fluid convection on Earth leads to large scale inhomogeneities of temperature and alloy concentration in bulk samples. Those inhomogeneities make it difficult to probe the spatially extended pattern dynamics under well-controlled diffusive growth conditions with constant growth velocity, temperature gradient, and alloy composition. For this reason, basic experimental studies of interface dynamics have been restricted to thin samples where fluid flow is eliminated. In addition, computational modeling of 3D patterns has remained challenging due to the several orders of magnitude disparity between capillary and transport scales. Detailed experimental studies in a microgravity environment where (i) convection is reduced sufficiently to obtain growth under diffusive conditions and (ii) the interface evolution can be directly visualized in three dimensions provide an unprecedented opportunity to elucidate fundamental mechanisms of the formation of those hierarchical structures by providing quantitative benchmark data that can be compared to predictions of computational models. The goal of this work is therefore to (i) obtain extensive benchmark experimental data under microgravity conditions over a range of velocity, thermal gradient, and alloy composition in a transparent system by imaging the dynamic evolution of microstructures under directional solidification conditions in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) installed onboard the International Space Station (ISS), and (ii) use state-of-the-art phase-field simulations to make quantitative predictions of dendritic hierarchical structure formation under diffusive growth conditions that can be directly compared to experimental observations under prescribed growth conditions. We expect the combination of those experimental and computational studies to establish a reliable framework for predicting dendritic microstructures in a wide range of alloy systems of technological relevance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. By bringing together well-controlled and accurately measured three-dimensional solidification experiments with quantitative phase-field simulations, this research should bring significant progress in understanding the origin of side-branching and the nature of the cell to dendrite transition, which have been a long standing issue in solidification for several decades. The deeper understanding of microstructure formation dynamics shall thus accelerate and optimize the design of new technological alloys and their processing conditions that can directly benefit US industries.
	During the past year no experiments were carried out on the International Space Station (ISS), so that the research was concentrated on analyzing some of the data obtained in the DSI project that needed quantitative analysis. Also, ground-based experiments were carried out in the DECLIC facility to compare the microstructures that develop under microgravity conditions on the International Space Station and under terrestrial conditions where significant gravity effects are present. In addition, ground-based experiments were also carried out in thin samples to obtain critical information required in planning and analyzing microgravity experiments. I. Analysis of Microgravity experimental data. Our previous microgravity experiments in the cellular regime revealed a rich variety of 3D breathing modes with limited spatiotemporal coherence. At low growth velocities and high enough temperature gradient, cellular arrays exhibit a global oscillatory behavior with an oscillation period homogeneous over the spatially extended array. However, phase locking was found not to occur throughout the whole system, but only very locally among neighboring cells. Such behaviors were reproduced with phase-field simulations with a remarkable accuracy on the predicted oscillation period (e.g., experiments: t ˜ 45.6 min, phase-field: t ˜ 44.6 min, at G=28 K/cm and V=1 μm/s). Moreover, simulations clarified the underlying physics of this oscillatory behavior: (i) the oscillatory behavior occurs at high temperature gradients at the end of a stable spacing branch when a gap in the stable array spacing appears; (ii) the lack of global coherence in experiments is due to the spatial reorganization of cells, maintained notably by tip splitting events; (iii) breathing mode oscillations were present over a narrow range of growth velocities for control parameters that do not lead to gap in the stable spacing branch. The identification of this subtle oscillatory mechanism constitutes the first quantitative agreement between observed and calculated dynamics in a three-dimensional configuration. As reported previously [1], unique observations were made for the first time in a fully three-dimensional configuration in SCN - 0.24 wt% camphor, including: (i) unique oscillating patterns of two-dimentional cellular arrays at high temperature gradient (G = 28 K/cm) and low velocities (V = 0.5 to 1.5 μm/s); (ii) secondary instabilities and growth of multiplet structures at low temperature gradient (G = 12 K/cm) and low velocities (V = 0.25 to 0.35 μm/s); (iii) homogeneous extended dendritic arrays showing coherent sidebranch formation at high velocities (V = 20 μm/s) and a wide distribution of primary spacing. We carried out a detailed analysis of the experimentally observed oscillatory modes of three-dimensional cellular patterns at different velocity conditions to complete our previous observations at one velocity. These experimental results were then compared with the additional phase-field simulations. During the past year we analyzed the observations from International Space Station to characterize the effect of growth parameters, crystal orientation, and sample history on promoting oscillations. It was found that cellular patterns displayed a remarkably uniform oscillation period throughout the entire array, despite the presence of high array disorder and the drifting of the array pattern. Oscillations were observed only for one velocity for G = 12 K/cm, but oscillations at different velocities were observed for G = 19 K/cm. This limited values of the gradient prevented quantitative correlation between the period of oscillation and the thermal gradient. However, the experimental period as a function of V can be obtained for the thermal gradient, G = 19 K/cm. The period of oscillation is found to have the power law relationship:t = 2.8 x 103 V-1.5, with t in seconds and V in μm/s. Oscillatory patterns were observed over a narrow range of parameters. In our studies, oscillations were present when 2 < V/Vc < 6, where Vc is the critical velocity for planar interface instability. Close to the higher limit of this velocity range, oscillations were found to occur only during the transient growth. One of the interesting observations was the presence of a drift of the pattern when the preferred crystallographic direction was not aligned with the pulling direction. This pattern drift was found to influence oscillations. Since the misalignment of cells in an array becomes larger with velocity, a critical drifting velocity exists above which oscillations are inhibited. Experiments in which the growth rate

Task Progress:	was changed showed that no oscillations were present when the velocity was dropped from a higher value to a lower value where oscillations were observed when starting from rest. II. Comparison between the microgravity and terrestrial experiments in bulk samples. In order to study the dynamical formation of two-dimensional arrays of cells/dendrites under diffusive growth conditions, 3D-experiments of directional solidification were conducted in microgravity. After the return of DSI on Earth, selected microgravity experiments were repeated under gravity-driven fluid flow to characterize the influence of convection on microstructure formation. The front recoils, the microstructure homogeneity along the interface, and the evolution of primary spacing were examined for both conditions. Initially, critical velocity was determined using the top view to detect the instability and the corresponding velocity was characterized from the side view observations. Onboard ISS, the critical velocity was found to be 0.48 µm/s, while on ground it was 0.95 µm/s. The larger critical velocity on ground indicates the presence of mixing in the liquid due to convection that reduces the concentration gradient at the interface. Under microgravity, patterns were observed to be more homogeneous than on ground where a very clear radial variation of size going from large cells in the center to small cells at the crucible border is observed. Comparative analyses of microgravity and ground experiments concerning the primary spacing evolution with pulling rate showed major differences. Similar primary spacing values are found for a macroscopically flat interface (2 µm/s) as the convection effect is weak. For higher pulling rates, spacing under microgravity conditions was smaller than that on the ground. III. The effect of sample size on convection and microstructure formation. Previously we had discussed that two different phase diagrams have been proposed in the literature that have significantly different solidus lines, and we had shown that both these solidus lines are not accurate. In one study the solidus line was obtained from directional solidification experiments in which the solidus temperature is equal to the planar interface temperature under steady-state diffusive growth condition. In general, diffusive growth is obtained in thin samples, and convection effects become negligible either in thinner samples or at higher growth velocities. To obtain stable planar front growth in succdinonirile-0.24 wr% camphor, a significantly low velocity is required that can induce convection if the sample size is not sufficiently small. We had shown that convection effects are not negligible in the sample thickness of 200 µm at V = 0.3 µm/s and sample thickness of 100 µm is required to obtain diffusive growth. Experiments in 100 µm thick samples were carried out to characterize the solidus line in the dilute composition range. In order to characterize convection effects as a function of sample thickness, detailed and systematic directional solidification experiments we carried out as a function of sample thickness, detailed and systematic directional solidification experiments we carried out as a function of sample thickness. Variations in interface temperature with time for different thicknesses, with V = 0.3 µm/s and G = 78 K/cm, were carried out over along time that was larger than 30 h. Thermal diffusivity in organic alloys is several orders of magnitude larger than the solute diffusivity, which means that heat diffusion is much faster than solute diffusion. Thus, slow solute diffusion is often the limiting physical process and the solute diffusion length ls = DL/V is the most important scale. In the time scale, this length corresponds to ts = DL/V². For SCN - 0.24 wt% camphor and V = 0.3 µm/s, ls = 900 µm and ts = 0.83 h which means that 30 h of plateau is more than 30 times higher than this limiting time scale. Experimental results were analyzed by using the boundary later model for transient planar front growth. The value of the dimensionless boundary layer thickness, Delta, was obtained for each thickness by matching the experimental interface temperature with the predictions of the model. The limiting case of diffusive growth occurs when Delta goes to infinity or the diffusive growth is approached when Delta = 7.8. Experimental results on the variation in Delta with thickness showed that diffusive growth is present only when the sample thickness is 100 µm or less. For samples larger than 100 µm, convection effects were present in the experiment under experimental values of velocity and thermal gradient. IV. The effect of imposed oscillatory velocity on sidebranch formation. In the dendritic regime, experiments on a SCN - 0.24 wt% camphor alloy suggested the existence of possible coherent modes of dendritic sidebranching for high growth velocities. This mechanism of coherent side branching was earlier reported in thin sample experiments on a dilute SCN alloy as well as in 2D and thin-sample 3D phase-field simulations, showing that this mechanism of coherent branching events is related to a limit cycle. We theoretically confirmed the existence of this coherent branching mechanism with full 3D phase-field simulations. We are currently investigating the link between the spatial order of spatially extended dendritic arrays and their dynamics of growth and sidebranching. One of the goals of this research project is to establish the mechanism of side branch formation. Two different theories now exist. The first model assumes that the onset of sidebranch is due to random thermal noise at the tip that amplifies to form sidebranches as it propagates along the sides. The side branches that form along the different sides of a given dendrite are not correlated when their formation is due to the thermal noise. Another mechanism for side branch formation is due to the oscillation n the tip and the tip radius that are correlated to the side branch evolution and in this case different sidebranches of a dendrite are correlated. We propose to carry out experiments in which coherent side branches can be induced by designing a sinusoidal velocity in the sample translation. We first carry out constant velocity experiments and once a steady state is reached, we switch over to a sinusoidal velocity and vary the amplitude and frequency of imposed oscillations to characterize their effects on the amplitude of sidebranches. Reference [1] N. Bergeon, D. Tourret, L. Chen, J. M. Debierre, R. Guerin, A. Ramirez, B. Billia, A. Karma, and R. Trivedi, Phys. Rev. Lett. 110, 226102 (2013).
Bibliography Type:	Description: (Last Updated: 04/03/2024)
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Awards	Trivedi R. "Principal Investigator Rohit Trivedi was awarded the Honorary Membership in The Materials Society (TMS), February 2017." Feb-2017